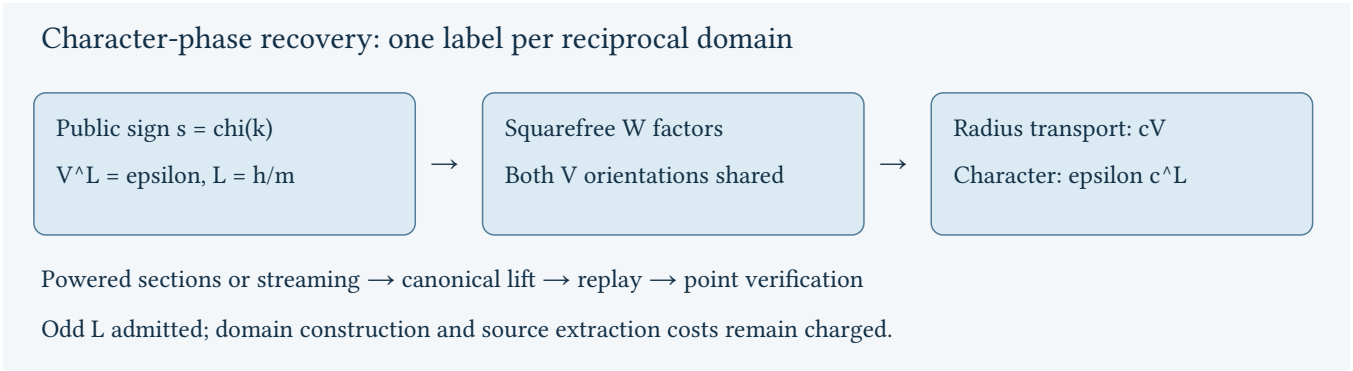
Character-phase canonical recovery

T-1918 · Prime fields · 2026-11-08

The new adapter uses the initial public character directly as its phase label. It admits m dividing h=(r−1)/2, including odd L=h/m, without constructing a fourth root of −1. The source contract and final point verification remain explicit.



Invariant. With $x=k/b_0$ and $V=x^m$, the phase is $V^L=\varepsilon=\chi(k)\chi(b_0)$. In $W=V+V^{-1}$ the domain is the squarefree part of $D_L-2\varepsilon$. Inversion preserves this character. Canonical fibers retain both orientations together, separate ramification at $W=\pm2$, and filter any orientation subsequently distinguished by observations.

Transport. At another radius, the variable is cV and its character is εc^L. The streaming signed equation retains $(-1)^h$, which matters for odd h. The powered path directly evaluates the signed mode product. Both paths impose mode0 at initialization.

Qualification. 1,936 factor controls and 23,552 direct-sign support equalities pass. All 3,600 public cases recover correctly, including 228 cases on the newly admitted order-19, m=3 fixture. The 3,372 predecessor public cases retain outcomes, routes and recorded source statistics. 298 single-call budget cases pass. PCC-20261108-cfa48d is unsigned producer evidence; independent review pending.

Custody and failures. Validation01 failed to compile at a test tuple equality: PullStats lacks PartialEq. No tests ran. The snapshot and compiler output are retained. A fresh validation02 binary passes focused and inherited checks; 131 copied sources and 146 compiler dependencies are hash-bound. Inherited output is byte-identical. Limits were 90/60/60 seconds. No timing ranking or cloud run.

Costs and next step. Domains still grow with L. All canonical checks, phase powers, inversions, allocations, source extraction, replay and point checks remain charged. The section-slot cap is per branch, not peak memory. Next reduce initial phase-domain work while preserving the exact character and transport invariant.

Sources. Predecessor reciprocal/canonical/signed/transport modules and NIST DLMF 18.5.1–2. The derivation and receipt list exact local sources. KB unavailable; no novelty claim.